PredMainAI — Phase 3 MVP Report

Group 3 - MAE 301
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Demo Video

1. Executive Summary

Industrial machines often fail unexpectedly, leading to costly downtime and maintenance. Traditional maintenance approaches either react too late or rely on rigid rule-based thresholds that fail to capture complex failure patterns.

PredMain AI is a predictive maintenance system that analyzes machine sensor data to detect anomalies and predict failures before they occur. The MVP allows users to upload a dataset of sensor readings and receive:

- Failure predictions across multiple models
- Machine health scores
- Risk rankings of machines
- Visual insights into failure drivers

The system combines anomaly detection (Isolation Forest, Autoencoder) with sequence modeling (LSTM) to provide both detection and interpretability.

2. User & Use Case

Target User

- Maintenance engineer in a manufacturing plant

Use Case Scenario

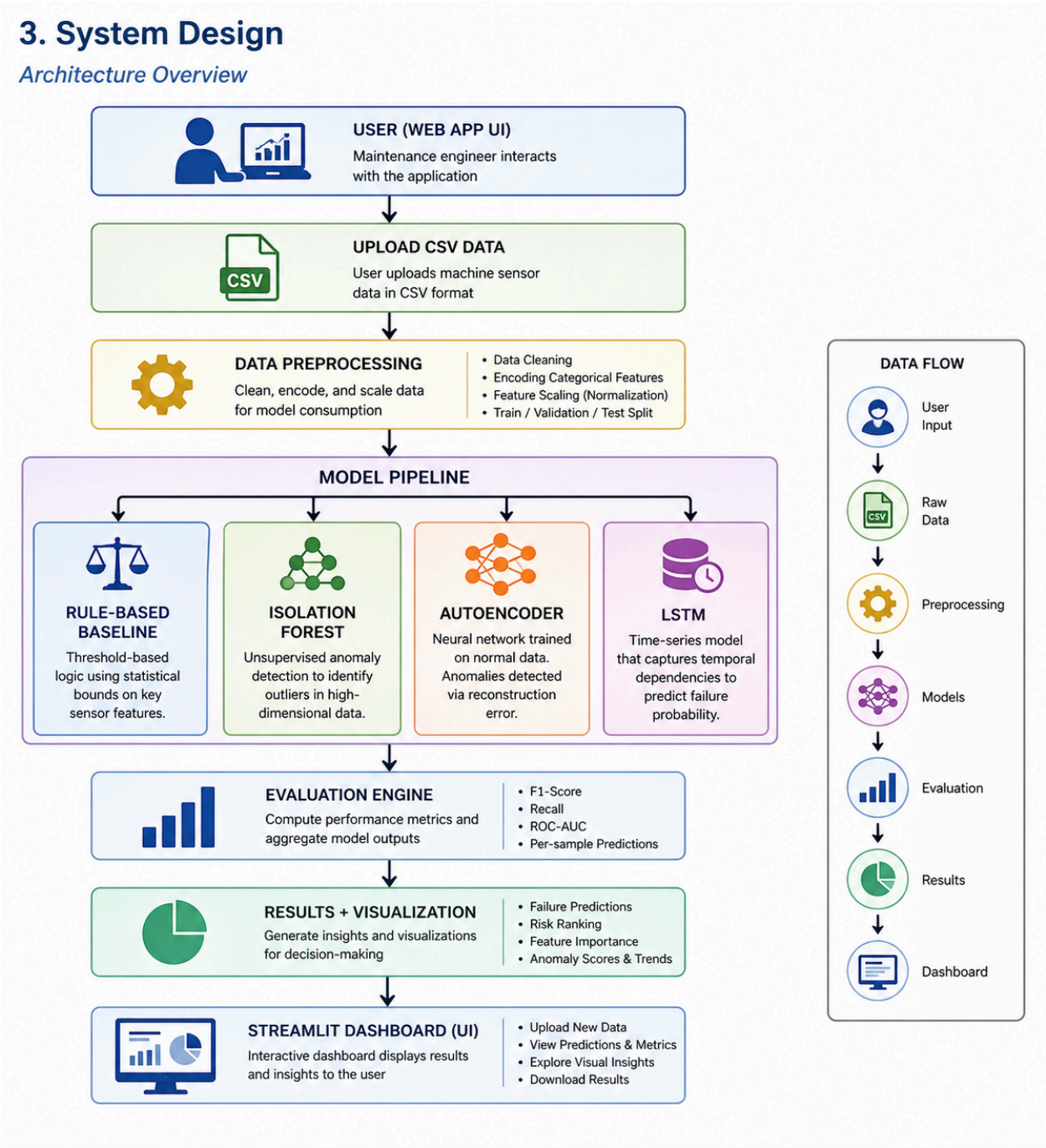
A maintenance engineer uploads a CSV file containing machine sensor data (temperature, torque, wear, etc.). The system processes the data and outputs:

- Which machines are likely to fail
- A ranked list of high-risk machines
- Explanations of why failures are predicted
- Visual insights into system behavior

This allows the engineer to:

- prioritize maintenance
- reduce downtime
- act proactively instead of reactively

3. System Design



Key Design Points

- Streamlit serves as the frontend interface
- Models are executed in a unified pipeline
- Results are aggregated into a single DataFrame
- Outputs include both predictions and explanations

4. Data

Dataset Used

- AI4I 2020 Predictive Maintenance Dataset (UCI)

Size

- ~10,000 samples
- ~9 features

Features

- Air temperature
- Process temperature
- Rotational speed
- Torque
- Tool wear
- Machine type

Preprocessing Steps

- Renamed columns to standardized format
- Encoded categorical variables (Type → numeric)
- Normalized numerical features (StandardScaler)
- Created binary target (failure_label)

Data Split

- Train: 70%
- Validation: 15%
- Test: 15%
- Stratified by failure class

5. Models

The system uses a combination of baseline and AI models:

Rule-Based Baseline

- Threshold-based detection
- Flags failure when:
    - torque or speed exceeds statistical bounds

Isolation Forest

- Unsupervised anomaly detection
- Trained only on normal samples
- Detects deviations from typical machine behavior

Autoencoder (Neural Network)

- Learns compressed representation of normal data
- Flags anomalies based on reconstruction error

LSTM (Time-Series Model)

- Uses sliding window of sensor data
- Captures temporal patterns in machine behavior
- Predicts failure probability over time

Model Strategy

- Hybrid approach:
    - statistical baseline
    - unsupervised anomaly detection
    - deep learning model

This improves robustness across different failure patterns.

6. Evaluation

Quantitative Results (Example)

Model	F1 Score	Recall	ROC-AUC	Verdict
Isolation Forest	0.485	0.941	0.934	Decent
LSTM	0.381	0.235	0.923	Weak
Autoencoder	0.301	1.000	0.519	Weak
Rule-Based	0.296	0.235	0.611	Weak

Key Insights

- Isolation Forest performs best overall
- Autoencoder detects all failures but produces many false alarms
- LSTM underperforms due to limited temporal structure in data
- High tool wear strongly correlates with failures
- Elevated torque + wear is a key failure pattern
- Some false positives show early warning signals rather than errors

7. Limitations & Risks

Technical Limitations

- LSTM performance limited by lack of strong time-series structure
- High false positive rate in anomaly models
- Duplicate column issues required post-processing fixes

Data Limitations

- Class imbalance (~97% normal)
- Synthetic-like dataset may not reflect real industrial variability

Risks

- False alarms → unnecessary maintenance
- Missed failures → critical system breakdown
- Model generalization may be limited to similar datasets

8. Next Steps

Technical Improvements

- Improve LSTM with better sequence engineering
- Tune anomaly thresholds to reduce false positives
- Add feature importance (e.g., SHAP values)

Product Improvements

- Deploy fully on Streamlit Cloud
- Add real-time monitoring dashboard
- Support live sensor streaming

Advanced Expansion

- Integrate with IoT systems
- Add predictive scheduling (when failure will occur)
- Build API for industrial integration

Summary

PredMain AI successfully demonstrates a working predictive maintenance MVP that:

- detects failures
- ranks machine risk
- provides interpretable insights

While performance can be improved, the system delivers a strong foundation for real-world deployment and further development.